

LECTURES

Cristian Giardinà - Integrable models of heat conduction

n/a

Lorenzo Piroli - Exactly solvable models of random quantum circuits

n/a

SEMINARS

Davide Lai - Clustering and the Five-Point Function

The hexagonalisation technique arose in the context of $N=4$ SYM theory to address the tessellation of the effective world-sheet describing three-point functions of single trace operators.

Despite its apparent technical difficulty, exploiting the knowledge of all its ingredients at finite 't Hooft coupling one can perform some non-trivial limits, like the strong-coupling one. In this case, a phenomenon called "clustering" allows the complete resummation of finite-size effects leading to the semi-classical string answer. I will introduce the general background and then discuss the generalization to some particular class of five-point functions, where the clustering phenomenon becomes more involved, involving cross-ratios and some analytic continuation of the ingredients to unusual kinematic regimes. In the end, I will speculate about some tentative directions that one could take.

Pavel Orlov - On diagonalization of quantum KdV hierarchy at large central charge

In this talk, I will introduce the quantum KdV hierarchy, one of the most well-known integrable structures in conformal field theory. Focusing on the large central charge limit, I will demonstrate how the eigenvectors and eigenvalues of the associated conserved charges can be systematically obtained through a perturbative expansion in $1/c$. The eigenvectors are constructed using the framework of the Adiabatic Gauge Potential, while the eigenvalues are derived from perturbative solutions of the Bethe ansatz equations. At the end I will discuss possible applications of these results.

Filiberto Ares - Entanglement asymmetry and the quantum Mpemba effect

The Mpemba effect is the counterintuitive and controversial phenomenon that hot water cools faster than cold one, or in more formal words, the more a system is out of equilibrium, the faster it relaxes. We introduce an analogous effect in extended quantum systems, in which a symmetry explicitly broken by the initial state is dynamically restored by the time evolution after a quantum quench. Unexpectedly, we find that the more the symmetry is initially broken, the faster is restored. To study this phenomenon, we borrow methods from the theory of entanglement in many-body systems and we introduce a quantity, dubbed entanglement asymmetry, which measures how much a symmetry is broken in a subsystem. In contrast to the classical case, we can establish the criteria for its occurrence in arbitrary

integrable quantum systems and provide direct experimental evidence using a trapped-ion quantum simulator.

Niels Benedikter - Derivation of the Luttinger Model from the Interacting Fermi Gas

The Luttinger model is famous as a model of interacting fermionic quantum particles in one space dimension due to the fact that it can be exactly solved using the bosonization method discovered by Mattis and Lieb in 1965. This method however is very specific to the model with its linear dispersion relation and its particular form of interaction. I will discuss how the Luttinger model however arises in a mean-field limit of a general fermionic quantum many-body model which is not exactly solvable.

Francesco Ravanini - Hagedorn singularity in exact $U_q(\mathfrak{su}(2))$ S-matrix theories with arbitrary spins

Generalizing the quantum sine-Gordon and sausage models, we construct exact S-matrices for higher spin representations with quantum $U_q(\mathfrak{su}(2))$ symmetry, which satisfy unitarity, crossing-symmetry, and the Yang-Baxter equations with minimality assumption, i.e. without any unnecessary CDD factor. The deformation parameter q is related to a coupling constant. Based on these S-matrices, we derive the thermodynamic Bethe ansatz equations for q a root of unity in terms of a universal kernel where the nodes are connected by graphs of non-Dynkin type. We solve these equations numerically to find out Hagedorn-like singularities in the free energies at some critical scales and find a universality in the critical exponents, all near 0.5 for different values of the spin and the coupling constant.

Henrik Juergens - About the structure of the correlation functions of the $sl(n+1)$ invariant model

n/a

Andrea Appel - n/a

Davide Fioravanti - n/a

Marco Meineri - n/a

Lenart Zadnik - n/a

Francesco Buccheri - n/a

Saverio Rota - n/a